Requirements / Specification Document

NIST
National Institute of Standards and Technology
U.S. Department of Commerce

TITLE: QNet Qubit Controller

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Radio Frequency Technology Division
Highspeed Waveform Metrology Group (672.03)

General Statement of Need:

The product to be procured is a low noise, programmable, RF and microwave pulse generator with capability to generate and synchronize multiple tones between DC and nominally 16 GHz. This equipment will be essential for the operation and measurement of microwave-optical transducer devices being developed at NIST and by collaborators at the University of Colorado and Joint Institute for Laboratory Astrophysics (CU-JILA). As part of the quantum networking project within the Communications Technology Laboratory (CTL), these transducer devices will be installed at NIST, with the purchased qubit controller acting as the experimental brain driving a sequence of pulses to different devices involved in the experiment to achieve measurements of quantum operation. Although these transducer devices have traditionally been operated in the continuous variable (CV) regime of quantum information, future results dependent on using pulses for discreet variable (DV) schemes for quantum measurement have recently been showing great promise. Precise adjustment of the timing between control pulses across many different frequencies and phases is a key performance requirement of these control systems. The qubit controller to be acquired by NIST must have high quality oscillator performance at each of the device frequencies, while also possessing the capability to quickly pulse in sequence each of the various output channels such that time dependent transducer experiments can be easily designed and implemented.

There are two critical technical requirements, namely multiple programmable output channels across the needed frequency ranges for driving different portions of the experimental apparatus and synchronization of internal oscillators for input heterodyne detection channels. Each output channel for the device must also possess high enough output power to ensure the generated signals can be amplified to correctly operate high input power optical modulators or to overcome the attenuation of the dilution refrigerator for driving microwave. Indeed, given the broad range of frequencies for transducer devices in the microwave frequency domain, and the need to drive both microwave devices and optical devices, a wide tuning range from nearly DC-16 GHz is required for each output channel. Another technical requirement is the ability to control and program this equipment so that it can set complete experimental sequences to gather results from

Additional specifications are noted below.

Item Specifications and Quantities:

Item 1:	QNet Qubit Controller	
Quantity:	1	
Specifications:	1.1	Output frequency tuning range: DC-16 GHz
	1.2	Multiple input/output channels with synchronized oscillators
	1.3	Output power: > +5dBm across the whole range
	1.4	Phase noise: < -145 dBc/Hz at 1 MHz offset from each oscillator
	1.5	Rack mount or bench top unit
	1.6	Manual and programmable control interface
	1.7	N-Type or 2.92 mm (K) coaxial output connector

Procurement Support Information and Ancillary Services:

Section		Requirements/Specifications
Planning Considerations ☒ N/A	☐	List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.]
	☐	US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.)
	☐	Vendor Representative On-Site more than three (3) days.
	☐	Other Considerations: [List other considerations to ensure an accurate quote.]
Shipping & Delivery ☐ N/A		Shipping Address:
	☒	NIST-Boulder Mail Stop: 672.03 325 Broadway Boulder, CO 80305
	☐	NIST-Gaithersburg ATTN: Mail Stop: 100 Bureau Dr Gaithersburg MD 20899
	☐	[Alternate Shipping Address]
		Shipping Criteria:
	☐	Partial Delivery Acceptable
	☐	[Enter Shipping and Delivery information by item.]
	☐	Direct Delivery to OU Building Required [Explain Need for Requirement] Building /Room Number: [Enter Building/Room Number]

☐ Building has Loading Dock
Containerization Preference: [Enter Container Type]
Other Requirements:
[List Any Other Requirement]
Delivery Date Criteria:
☒ Delivery Shall Be Competed No Later Than:
Click or tap to enter a date.
☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.
☐ Scheduled Deliveries:

Item Number	Due Date

☐ Other:
[List Other Shipping, Delivery or Special Requirements.]

Electronic Media
☒ N/A

[Enter Electronic Media Requirements by Item.]
☐ Software
☐ Electronic Manuals
☐ License Key
☐ Other:
[List Any Other Electronic Media Requirements]
☐ US Government Email Address for Software Delivery:
[Enter Address]

Installation
☒ N/A

[Enter Installation Requirements by Item]
☐ Vendor Technician on Site
☐ Rigging
☐ Uncrating / Unpackaging
☐ Removal of Packaging Material
☐ Equipment Set Up
☐ Start-Up Services
☐ Turnkey Installation
☐ Other:
[Enter Details]
NOTE: SME shall be the coordination POC.

Facility / Utility Considerations
☒ N/A

[List Site Specific Utility Considerations by Item]
☐ OFPM Work Order Numbers
[List Work Order Numbers]
☐ Power Available
Voltage: [Enter voltage] (V)
Maximum Current: [Enter max current draw] (A)
Phase: [Choose Phase]
Frequency: [Typically 60Hz] (Hz)
NEMA Plug Type: [Enter Plug Type] (i.e.: NEMA L5-20P)
Environment: [Choose Environment]
Other Power Considerations:
[List Any Other Power Considerations.]
☐ Utility Connections:
[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]
☐ Other Considerations:
[List Other Considerations to Ensure an Accurate Quote.]

Inspection and Acceptance
☐ N/A

[Inspection Requirements Per Item (Who, When, What, Where)]
☐ On-Site
☐ Acceptance Not Expected to Exceed 7 Business Days
☒ Acceptance Expected to Exceed 7 Business Days:
15 business days to conduct performance tests.
☐ Inspection and Acceptance Plan:
[Explanation]
☐ Other:
[Enter Other Requirements]

Training
☒ N/A

Training Requirements (Who, When, What, Where)
☐ Number of Trainees: [Enter Number of Trainees]
☐ On-Site

	☐ Off-Site Location: [Enter Location(s)] Is Trainee Travel Required: [Choose Answer] Scope of Training ☐ Operation ☐ Maintenance ☐ Troubleshooting ☐ Safety ☐ Other [Enter Other Scope of Training Requirements] ☐ Training Materials to be Provided by Government: [Describe Materials] ☐ Training Materials to be Provided by Vendor: [Describe Materials] ☐ Other [Enter Other Training Requirements] NOTE: SME shall be coordination POC.
Warranty ☐ N/A	Base Manufacturer Warranty is acceptable Warranty Length: [i.e. 3 - Years] What is included? ☐ Labor ☐ Parts ☐ Travel ☐ Cost and Liability for Returns ☐ Other [Enter Other Requirements] ☐ 1 Option Period ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Calibration ☒ N/A	[Describe Calibration Requirements Per Item] Calibration Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Maintenance ☒ N/A	[Describe Maintenance Requirements Per Item] Maintenance Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Government Furnished Material / Equipment ☒ N/A	List Any Government Furnished Material / Equipment Per Item] ☐ Samples for Testing ☐ Equipment which will leave Government Site Description: [Enter Description] Serial Number: [Enter Serial Number] NIST Property Number: [Enter NIST Property Number] ☐ Other: [Enter Other Details]
Travel ☒ N/A	[Describe any Travel Requirements Per Item] ☐ Explain: [Enter Justification / Requirement] ☐ Other:

[Enter Other Requirements]

NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.